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SUMMARY

Computer Engineering graduate focused on representation learning and deep generative models. I build and study autoencoders, VAEs, and latent-variable methods, with hands-on work across computer vision and vision-language. I aim to pursue thesis-based graduate research in multimodal representation learning, toward a PhD.

EDUCATION

Shahid Bahonar University of Kerman

Kerman, Iran

B.Sc. in Computer Engineering

Sep 2020 – Sep 2025

- **GPA:** 14.94 / 20 cumulative; **~17.0 / 20** across AI/ML & core-CS coursework (see *Selected Coursework*).
- **B.Sc. Thesis:** *Data-driven estimation of intrinsic dimensionality for autoencoder architecture design* (Advisor: Dr. Mahdi Eftekhari).

RESEARCH INTERESTS

Multimodal representation learning; deep generative and latent-variable models (autoencoders, VAEs, normalizing flows, diffusion); representation compression and intrinsic-dimensionality estimation; self-supervised & contrastive learning (CLIP, DINO, MoCo) for vision-language tasks.

RESEARCH EXPERIENCE

Independent Research — Autoencoder Compression & Intrinsic Dimensionality

Self-directed continuation of prior thesis work

June 2026 – present

- **Core idea:** using unsupervised intrinsic-dimensionality estimates to size autoencoder bottlenecks under a target reconstruction-loss threshold.
- **Implementation:** comparing CNN-AEs, MLP-AEs, VAEs, and ResMLP models on MNIST, FashionMNIST, and CIFAR-10 (PyTorch). *Writeup in preparation.*

Research Assistant — Machine Learning Group

Kerman, Iran

Shahid Bahonar University of Kerman | Advisor: Dr. Mahdi Eftekhari

Aug 2024 – Sep 2025

- **Representation learning:** studied autoencoders, non-negative matrix factorization (NMF), and basis-driven analysis to estimate intrinsic dimensionality.
- **Architectures:** implemented enhanced autoencoders in PyTorch; this work became my B.Sc. thesis and a manuscript now being revised independently.

PROJECTS

Multi-Condition Synthetic Driving Dataset for Instance Segmentation

collaborative

CARLA + GroundedSAM annotation pipeline

May – June 2025

- **Dataset generation (Python, CARLA):** scripted day/night/rain/fog conditions and a synchronized dual-camera rig (RGB + instance-seg); converted contours to YOLO polygons — 3,704 frame pairs (~10.2K/1.9K/2.2K instances across vehicles, pedestrians, buses).
- **Auto-labeling (GroundedSAM):** grounded-prompt → SAM-mask pipeline to refine weak simulator ground truth; added position-based splits and Albumentations augmentation.
- **Handoff:** collaborator trained/evaluated YOLOv11-seg (n/m) and compiled to TensorRT FP16 (~4.1× faster inference). [repo](#)

Multi-View Attention for Steering-Angle Prediction

collaborative

ResMLP backbone + multi-camera attention

July – Aug 2025

- **Model design (PyTorch, ResMLP):** proposed and built a ResMLP backbone fused with a multi-view attention mechanism over center/left/right views, plus the data & training pipeline.
- **Scope:** evaluation suite (directional/angular metrics, attention-entropy analysis, bootstrap tests) built by collaborator. [repo](#)

Image Captioning — Show and Tell (PyTorch): CNN-encoder / LSTM-decoder captioning with greedy & beam-search decoding, evaluated with BLEU. [repo](#)

Transfer-Learning Pipeline (PyTorch): pretrained a CNN on a 27k-image ImageNet subset and fine-tuned on CIFAR-100.

TEACHING EXPERIENCE

Sole Teaching Assistant — Graduate Deep Learning Course

Shahid Bahonar Univ. of Kerman

As an undergraduate, under the supervision of Dr. Mahdi Eftekhari

Feb 2025 – Sep 2025

- **Sole TA** for a **master's-level** deep-learning course. Independently designed the complete assignment suite, including an **image-captioning** project (*Show and Tell*), a **transfer-learning** project (ImageNet-subset pretraining → CIFAR fine-tuning), and a **parameter-efficient fine-tuning (PEFT)** project in which students fine-tuned a **4-bit quantized GPT-2** model on the **SQuAD** dataset for question answering using **LoRA**.
- **Delivered lectures** on self-supervised and vision-language learning, covering **CLIP**, **DINO**, and **MoCo**, and mentored graduate students in implementing and debugging deep neural networks.

Head TA — Artificial Intelligence (Sep 2023 – Jan 2024): led the TA team and designed a capstone **multi-agent** project (reflex / model-based / goal-based agents in a competitive grid-world).

INDUSTRY EXPERIENCE

Machine Learning Engineer, Fusion Food Tech / Artechly

Dec 2025 – May 2026

Built an AI voice agent for end-to-end order collection; designed order-extraction, verification, and secure payment pipelines.

NLP & AI Engineer, AARTAA

Jun 2025 – Sep 2025

Built transformer-based NLP services (FastAPI, WebSocket); deployed models to production with OAuth 2.0.

Machine Learning Engineer, Ava Pardaz Keyhan Kariman

Feb 2024 – Jul 2024

Fine-tuned T5 for NER and keyword extraction; built a RAG chatbot and NLP retrieval pipelines.

TECHNICAL SKILLS

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- **Deep Learning & Generative Models:** autoencoders, VAEs, ResMLP, representation learning, self-supervised & contrastive learning (CLIP, DINO, MoCo), CNNs, RNNs.
 - **Computer Vision & Multimodal:** vision-language grounding (Grounding DINO / SAM), image captioning, synthetic data generation (CARLA).
 - **NLP & LLMs:** transformers, foundation models, fine-tuning (T5), sequence modeling.
 - **Programming & Tools:** Python, SQL, Rust (basic); PyTorch, TensorFlow, Keras, Hugging Face, scikit-learn; Git, Linux, Docker.

SELECTED COURSEWORK

(grades out of 20) Fundamentals of Computational Intelligence (20) • Basics of Speech & Language Processing (17.58) • Fundamentals of Data Mining (17.75) • Computer Vision Basics (16.75) • Special Topics in AI (16.5) • Machine Learning (16.0) • AI & Expert Systems (16.2) • Eng. Probability & Statistics (16.5).

REFERENCES

Available upon request.

CERTIFICATIONS & LANGUAGES

Certifications: ML Specialization (Coursera); NLP Specialization (Coursera); PyTorch for Deep Learning (ZTM).

Languages: Persian (native). English — **TOEFL iBT: Overall 5.0/6 ($\approx 100/120$)**; Listening 6.0, Reading 5.0, Writing 5.0, Speaking 4.5.

INTERESTS

Confectionery and pastry; outdoor adventures — hiking and cycling; gatherings with friends.